

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 3.1: Acids and Bases — Teacher Reference (Pre-filled)

SUMMARY TABLE: CHEMISTRY GRADE 10	
Sub-Strand	Sub-Strand 3.1: Acids and Bases
Driving Question	DRIVING QUESTION: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Phenomenon: Why Do Lemon Juice, Vinegar, Baking Soda and Soap Behave So Differently?	We observed that lemon juice (pH 2.0) and vinegar (pH 2.8) have pH values below pure water (pH 7.0), while baking soda solution (pH 8.3) and soap solution (pH 10.0) are above 7. Lemon juice and vinegar taste sour and are safe in food, baking soda tastes bitter/salty and is safe in small amounts in food, and soap feels slippery and is harmful if swallowed.	We learned that products that look alike can be on different sides of 7 on the pH scale, and that their uses (vinegar for pickling, soap for cleaning, baking soda for mandazi, lemon juice for flavour and tea) seem to match their pH, taste and feel.	We can so far hypothesise that something different inside lemon juice and vinegar from inside baking soda and soap causes the difference in pH, taste and feel, and that antacids may help hyperacidity because they are on the other side of the scale from stomach acid. We still need evidence to test this idea.
Lesson 2 Sorting the Kitchen: Classifying Household Substances as Acids or Bases with Litmus and Red Cabbage	Record what you saw: the colour of red litmus, blue litmus and red cabbage extract in lemon juice, vinegar, water, baking soda solution and soap solution.	Write how you can sort a substance as an acid or a base using the pattern of colour changes: acids turn blue litmus red; bases turn red litmus blue; water does not change either.	Explain how the indicator results classify the four kitchen products, how they compare with your Lesson 1 hypotheses and pH values (2.0, 2.8, 7.0, 8.3, 10.0), and what you still want to find out about why they behave this way and how this might affect their uses.
Lesson 3 Natural and Laboratory Indicators: Building a Classification Table	We tested lemon juice, vinegar, water, baking soda solution and soap solution with red cabbage extract and universal indicator. Red cabbage: lemon juice red/pink, vinegar pink-red, water purple, baking soda blue-green, soap green-yellow. Universal indicator: lemon juice red-orange (pH 2), vinegar orange (pH 3), water green (7),	Indicators show whether a substance is acidic, neutral or basic. Natural indicators such as red cabbage extract and laboratory indicators such as litmus and universal indicator all give the same sorting, even though the colours differ. Universal indicator also shows how acidic or basic a substance is on the pH scale. Water is neutral (pH	Lemon juice (pH 2.0) and vinegar (pH 2.8) are acids; baking soda (pH 8.3) and soap (pH 10.0) are bases; water (pH 7.0) is neutral. Lemon juice is more acidic than vinegar, and soap is more basic than baking soda. Our hypotheses from Lesson 1 are supported by the indicator evidence and the pH values, and our model shows

	baking soda blue (8), soap blue-purple (10).	7.0).	more H^+ in the acids and more OH^- in the bases. We still need to explain why acids are sour and soap is slippery.
Lesson 4 Why is Lemon Juice Sour? Acids Release H^+ Ions in Water (Arrhenius Definition)	We observed that lemon juice (pH 2.0, $H^+ = 1 \times 10^{-2}$ mol/L) and vinegar (pH 2.8, H^+ about 1.6×10^{-3} mol/L) have much more H^+ than pure water (pH 7.0, $H^+ = 1 \times 10^{-7}$ mol/L), and in the simulation the acid molecules released H^+ (aq) ions when they dissolved in water. Lemon juice has about 6 times more H^+ than vinegar.	We learned that an acid is a substance that produces H^+ (aq) ions in water (Arrhenius definition), and that a lower pH means a higher H^+ concentration. Each pH unit is a 10-fold change in H^+ concentration.	We can explain that lemon juice and vinegar are sour, turn blue litmus red and have low pH because the acids (citric acid and ethanoic acid (acetic acid)) release H^+ (aq) ions in water; lemon juice is more sour than vinegar because it has about 6 times more H^+ . Stomach acid is also an acid with H^+ ions, which helps us start to understand why antacids are needed; we still need to learn what particles bases like baking soda and soap release.
Lesson 5 Arrhenius Definition (Bases and Alkalis): Why Does Soap Feel Slippery?	Red litmus turned blue in baking soda solution and soap solution, but stayed red in lemon juice and vinegar. Soap solution (pH 10.0) felt slippery, baking soda solution (pH 8.3) was only weakly basic, and soap has an OH^- concentration of 1×10^{-4} mol/L, 1000 times more than pure water (1×10^{-7} mol/L).	Bases that dissolve in water are called alkalis and they produce OH^- (aq) ions (Arrhenius definition). All alkalis are bases but not all bases dissolve in water. Soap has more OH^- ions than baking soda, so it has a higher pH.	Soap feels slippery because it is an alkali that releases OH^- (aq) ions in water, and these ions react with the oils on the skin. Baking soda gives fewer OH^- ions, so it is only weakly basic and safe in small amounts in mandazi. Our particle model now shows H^+ (aq) in lemon juice and vinegar and OH^- (aq) in baking soda and soap.
Lesson 6 Chemical Properties I: How Acids React with Metals and Carbonates	Vinegar mixed with baking soda fizzed, and the gas turned limewater milky. Lemon juice also fizzed with baking soda, but water with baking soda did not. Magnesium ribbon in dilute hydrochloric acid gave bubbles, and the gas burned with a squeaky pop. Magnesium in water showed no bubbles.	Acids react with reactive metals such as magnesium to give a salt and hydrogen gas, and with carbonates such as baking soda to give a salt, water and carbon dioxide. Hydrogen is tested with a lighted splint (squeaky pop) and carbon dioxide with limewater (turns milky). These reactions happen because of the H^+ (aq) ions in acids.	The fizzing in the Kisumu kitchen happens because vinegar and lemon juice are acids that release H^+ (aq) ions, which react with the carbonate-type baking soda to give carbon dioxide. Soap does not do this because it is a base. This shows that the acidic or basic nature of a substance decides how it behaves and is used, which helps us answer the driving question and prepares us to explain how antacids deal with hyperacidity.
Lesson 7 Chemical Properties II: Neutralisation and Antacids	When magnesium hydroxide (or baking soda) was added to dilute hydrochloric acid with universal indicator, the colour changed from red towards orange, yellow and green as the pH rose; adding water alone changed it very little. The stomach pH rises from about 1.5 to 2 to about 3.5 to 5 after an antacid.	Neutralisation is a reaction between an acid and a base that gives salt + water. In terms of ions, H^+ (aq) + OH^- (aq) gives H_2O (l). Antacids contain bases such as magnesium hydroxide or calcium carbonate.	Patients with hyperacidity are given antacids because the base neutralises some of the excess hydrochloric acid in the stomach, forming salt and water and raising the pH to a safer level, so the use of a base as medicine depends on its alkalinity (and, as we will see, on its strength and safety).
Lesson 8	With universal indicator, 0.1 M hydrochloric	Strength is about the degree of ionisation,	At equal concentration, HCl gives many

Strength of Acids and Bases and the pH Scale	acid showed pH 1 and 0.1 M ethanoic acid showed pH 3; 0.1 M sodium hydroxide showed pH 13 and 0.1 M ammonia showed pH 11. Lemon juice (pH 2.0) is lower than vinegar (pH 2.8) on the pH graph.	not concentration. A strong acid or base ionises almost completely, while a weak one ionises only partly. Each pH unit is a 10-fold change in H^+ concentration, so lemon juice has about 6 times more H^+ (aq) than vinegar.	more H^+ (aq) ions than ethanoic acid, so its pH is lower; NaOH gives more OH^- (aq) than ammonia, so its pH is higher. This helps explain why a weak base like magnesium hydroxide is a safer antacid for hyperacidity than a strong base, and it advances our answer that alkalinity, together with strength and safety, influences how products are used.
Lesson 9 Uses of Acids and Bases and Safety: Why Vinegar Pickles, Soap Cleans, Lime Heals Soil and Antacids Soothe	Vinegar (pH 2.8) is used by mama mboga to pickle kachumbari and onions; soap solution (pH 10.0) lifts grease; baking soda (pH 8.3) is used for mandazi; lime raises Kericho soil from pH about 4.5 to about 6.0 for maize; a magnesium hydroxide antacid raises stomach pH from about 1.5-2 to about 3.5-5 after spicy nyama choma.	The uses of acids and bases depend on their nature (H^+ (aq) or OH^- (aq)), their pH and their strength, and each use needs matching safety precautions: acids help preserve and flavour food, bases clean, bake and neutralise acids in soil and in the stomach, and not every acid or base is safe in the same way.	Alkalinity largely determines uses because bases neutralise acids (acid + base gives salt + water): lime neutralises H^+ (aq) in acidic soil, antacids neutralise excess hydrochloric acid in hyperacidity, and soap's OH^- (aq) helps remove grease, while strength and safety decide how they can be used.
Lesson 10 Consolidation: Answering the Driving Question – Alkalinity, Uses and Antacids	We observed that lemon juice (pH 2.0) and vinegar (pH 2.8) are acidic, sour and turn blue litmus red, while baking soda (pH 8.3) and soap (pH 10.0) are basic, bitter or slippery and turn red litmus blue. We also observed that acid + base reactions give salt + water, that stomach pH rises from about 1.5 to 2 to about 3.5 to 5 with an antacid, and that lime raises Kericho soil pH from about 4.5 to 6.0.	We learned that acids release H^+ (aq) ions and alkalis produce OH^- (aq) ions, that lower pH means more H^+ , that strength (degree of ionisation) differs from concentration, and that properties such as pH, strength and safety determine how acids and bases are used for pickling, cleaning, baking, farming and medicine.	We explain that alkalinity, together with strength and safety, largely determines the uses of substances, and that patients with hyperacidity are given antacids such as magnesium hydroxide because the base neutralises excess stomach hydrochloric acid (H^+ (aq) + OH^- (aq) gives H_2O , acid + base gives salt + water), raising the stomach pH to a safer level.

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